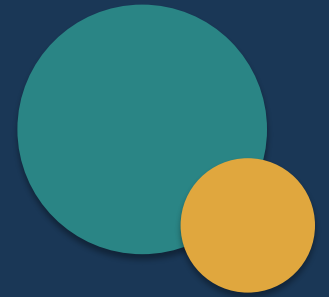


NUCLEOTIDE METABOLISM

DR. WAJEEHA SHAMS ANSARI

Purine Nucleotide Biosynthesis & Metabolism



Why are nucleotides important?

Nucleotides are much more than DNA/RNA building blocks.

Build genetic material

- RNA needs ribonucleotides.
- DNA needs deoxyribonucleotides.
- Nucleotides are the building blocks needed to make DNA and RNA. Without them, cells cannot grow or multiply.

Energy & signaling

- **ATP and GTP** provide **energy** for cell activities.
- cAMP and cGMP act as second messengers inside the cell.
- Nucleotides help regulate cell pathways.

Other roles

- Part of important coenzymes: FAD, NAD(H), NADP(H).
- They also form **activated molecules** such as **UDP-glucose** and **CDP-choline**, which help the cell make proteins, carbohydrates, fats, and other important molecules.

Big idea: cells make purines and pyrimidines either de novo (from small molecules) or by salvage (recycling preformed bases).

NOMENCLATURE

RIBONUCLEOTIDES:

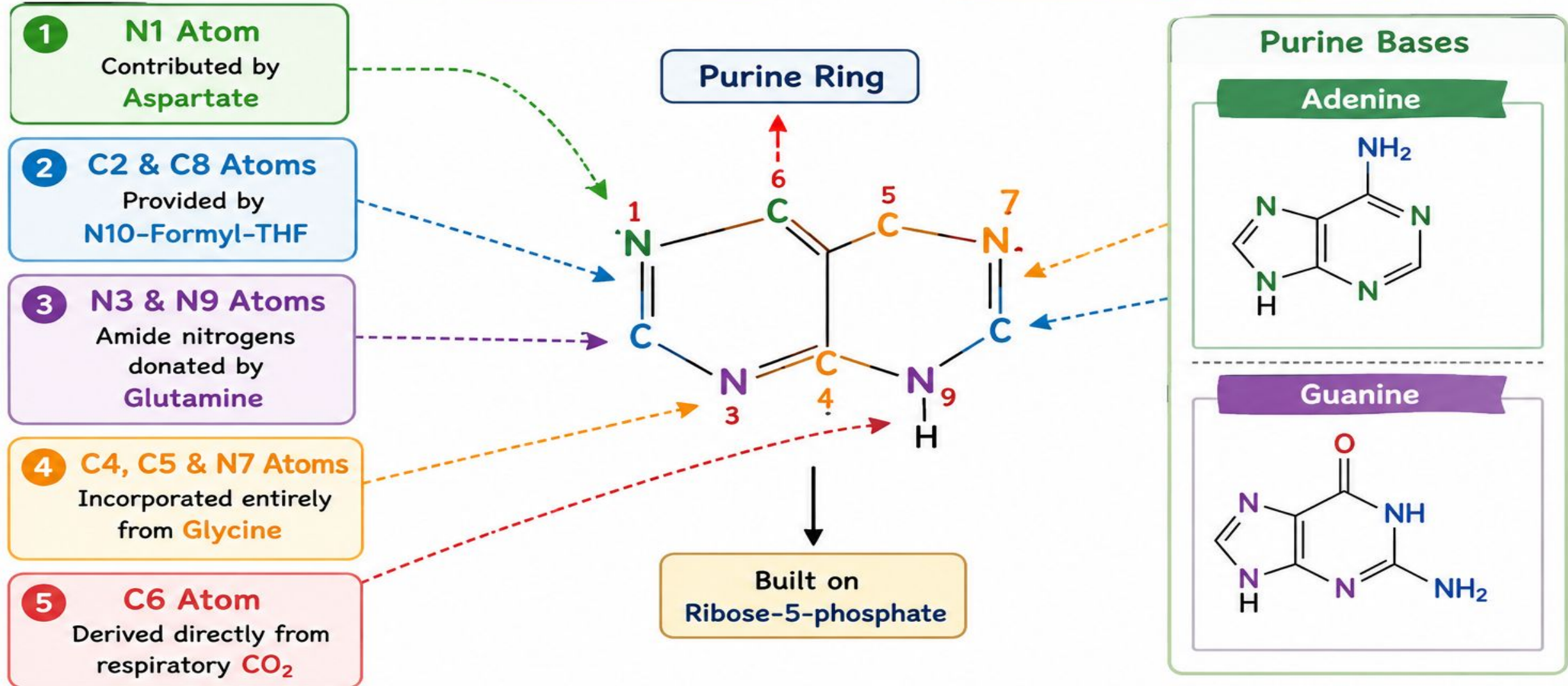
- ▣ ADENINE+ RIBOSE =ADENOSINE+ PHOSPHATE= ADENYLATE
- ▣ GUANINE+RIBOSE= GUANOSINE+PHOSPHATE=GUANYLATE
- ▣ CYTOSINE+RIBOSE=CYTIDINE+PHOSPHATE=CYTIDYLATE
- ▣ URACIL+RIBOSE=URIDINE+PHOSPHATE=URIDYLATE

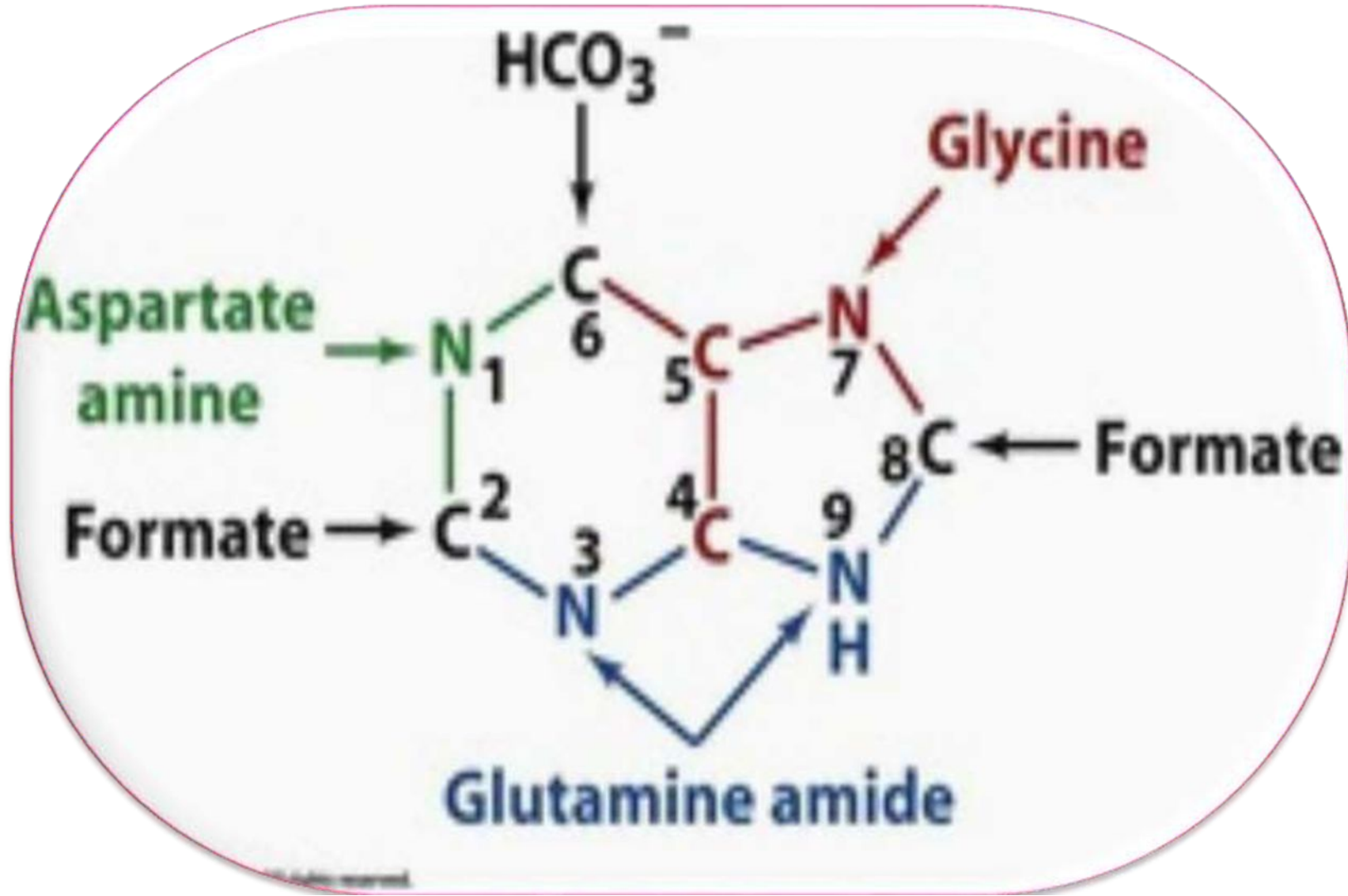
DEOXYRIBONUCLEOTIDES:

- ADENINE+ DEOXYRIBOSE =dADENOSINE+ PHOSPHATE= DEOXYADENYLATE OR dAMP
- GUANINE+dRIBOSE= dGUANOSINE+PHOSPHATE=Dguanylate or dGMP
- CYTOSINE+RIBOSE=CYTIDINE+PHOSPHATE=CYTIDYLATE
- THYMINE+RIBOSE=dTHYMIDINE+PHOSPHATE=
- THYMIDYLATE

Biosynthetic Origins of the Purine Ring

The 9-membered purine ring is built step-by-step on a pre-existing **ribose-5-phosphate** using small molecule precursors.





Tissue Distribution & Dual Metabolic Pathways



Primary Site: Liver

The **liver** is the primary organ for *de novo* purine synthesis. It expresses the full complement of cytosolic enzymes required to build purines from scratch and exports bases to peripheral tissues.



Dependent Tissues

Brain, Erythrocytes, and Leukocytes have low or absent *de novo* enzymatic capacity. They rely almost exclusively on salvage pathways or imported nucleosides to meet purine needs.

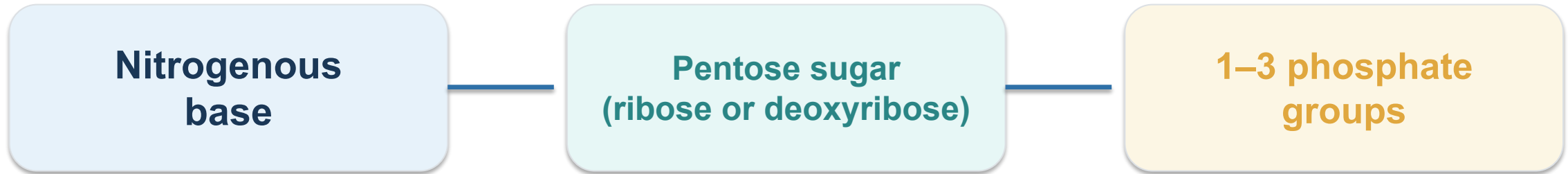


Salvage vs. De Novo

De novo purine synthesis uses a lot of energy—about **6 high-energy phosphate bonds** are needed to make one purine ring. In contrast, the **salvage pathway** reuses purine bases that are already present, so it **requires much less energy**.

Nucleotide structure: 3 parts

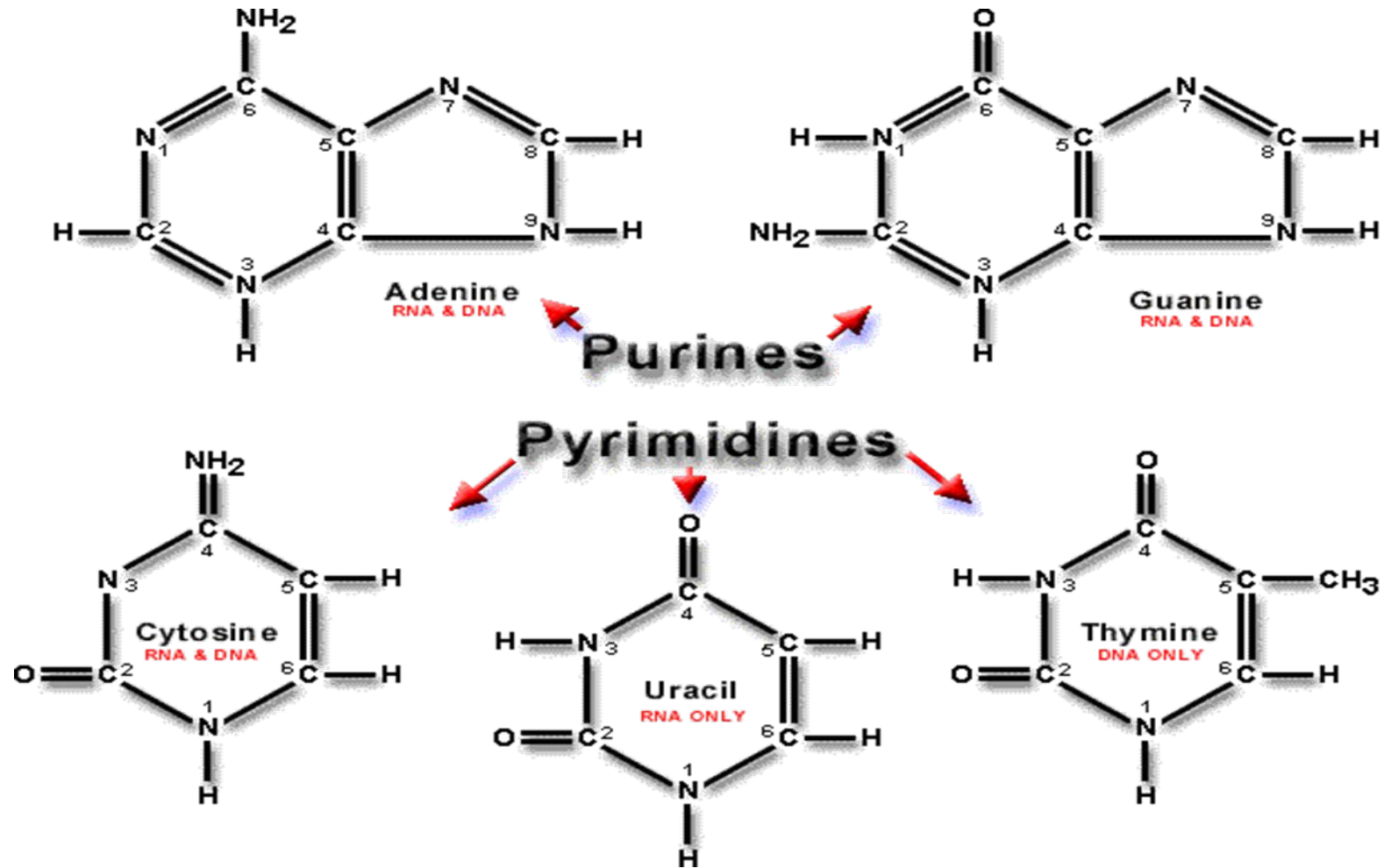
Think: base + sugar + phosphate(s).



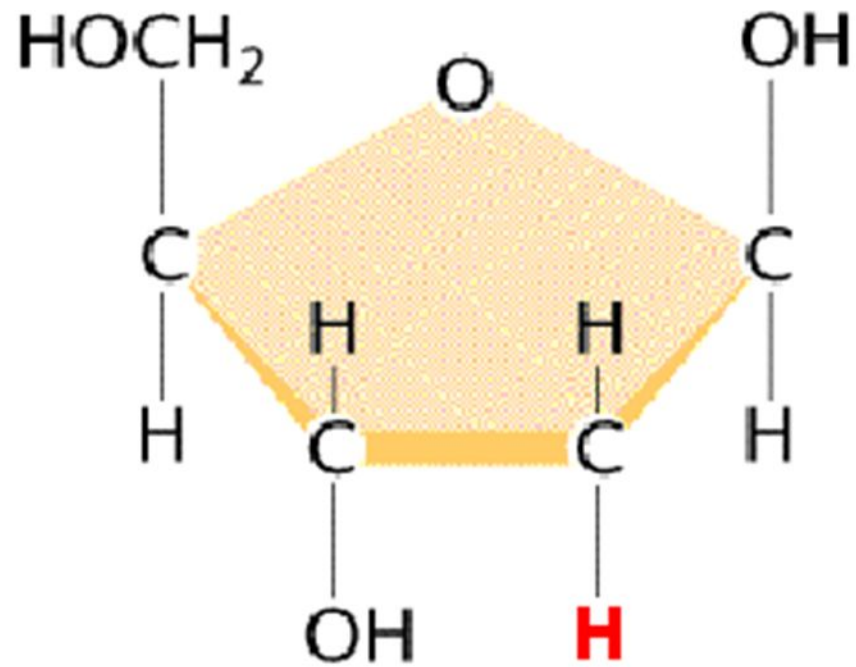
Nucleoside = base + sugar. Nucleotide = nucleoside + one or more phosphate groups.

- Purines: adenine (A) and guanine (G) — two-ring bases.
- Pyrimidines: cytosine (C), thymine (T), and uracil (U) — one-ring bases.
- DNA uses A, G, C, T. RNA uses A, G, C, U.

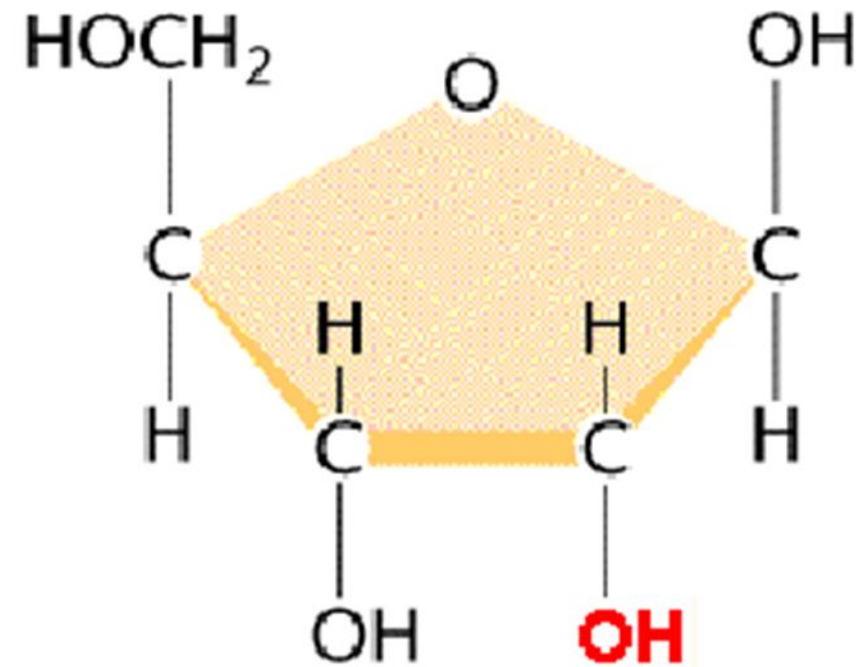
PURINE AND PYRIMIDINE STRUCTURE



PENTOSE SUGAR STRUCTURE



2-Deoxyribose



Ribose

Nucleosides vs nucleotides

The sugar determines whether it is a ribo- or deoxyribo- compound.

Nucleoside

- Base + pentose sugar
- Ribose → ribonucleoside
- 2-deoxyribose → deoxyribonucleoside
- Examples: adenosine, guanosine, cytidine, uridine

Nucleotide

- Nucleoside + phosphate(s)
- 1 phosphate → monophosphate (AMP)
- 2 phosphates → diphosphate (ADP)
- 3 phosphates → triphosphate (ATP)

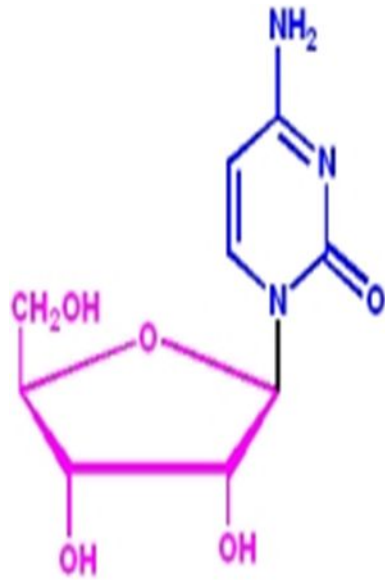
Adenosine

AMP

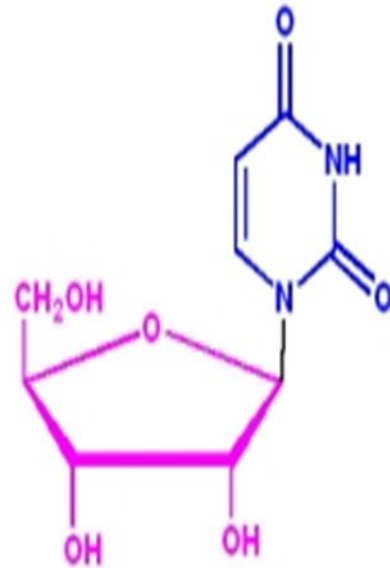
ADP → ATP

Phosphate groups give nucleotides negative charge. The second and third phosphates of NTPs(nucleoside triphosphates) are associated with high-energy bonds.

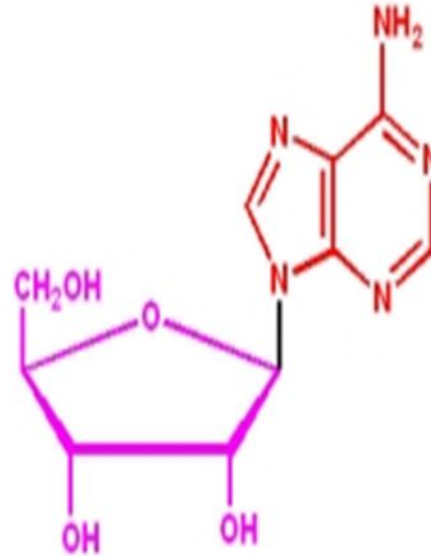
NUCLEOSIDES STRUCTURE



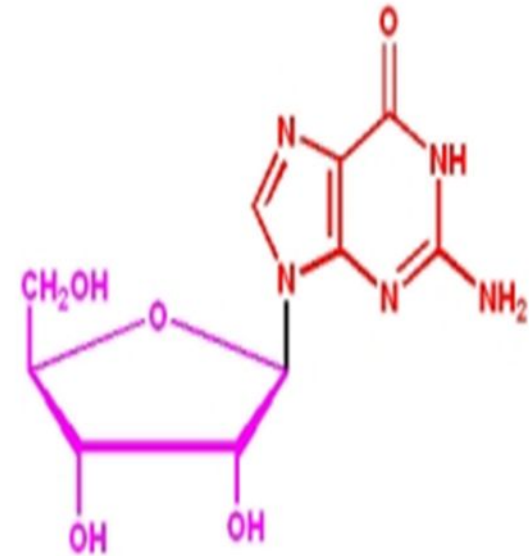
Cytidine



Uridine



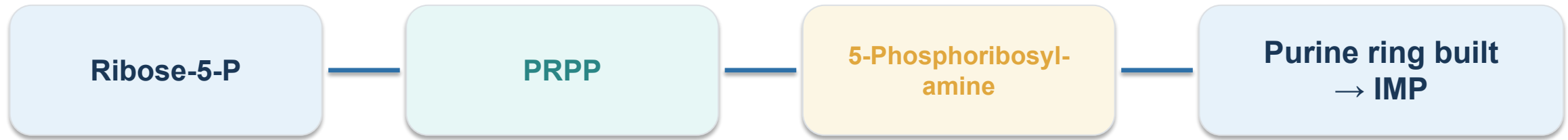
Adenosine



Guanosine

Purine nucleotide synthesis

The purine ring is built step-by-step on a ribose phosphate scaffold.

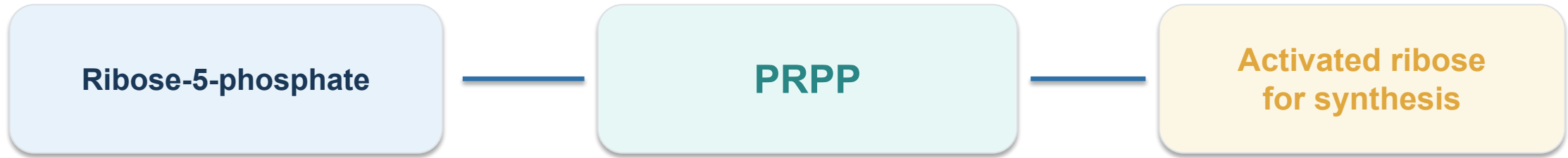


- Purine ring atoms come from glycine, glutamine, aspartate, CO₂, and N¹⁰-formyl-THF.
- ATP is required at several steps.
- IMP is the parent purine nucleotide → it branches to AMP or GMP.

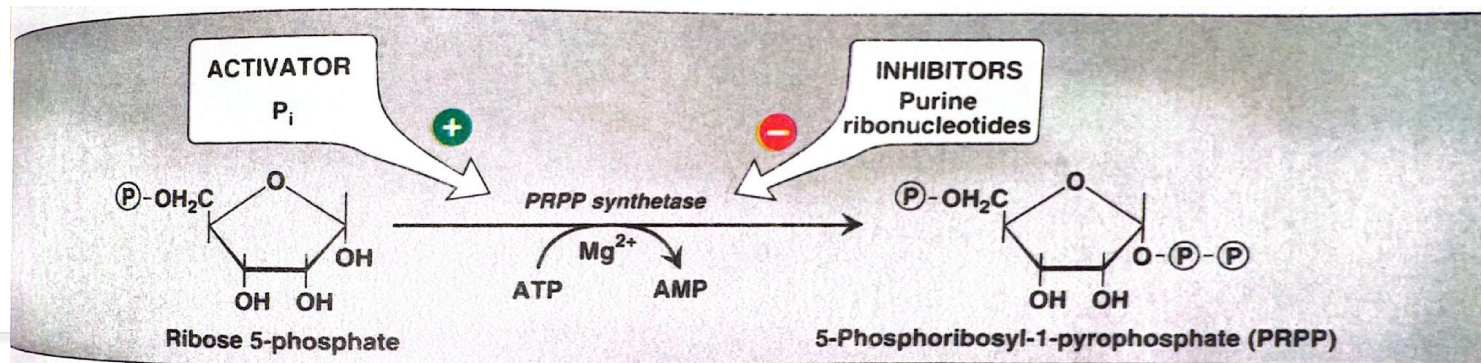
Key concept: unlike pyrimidines, the purine ring is assembled directly on the ribose phosphate.

Step 1: PRPP formation

PRPP = 5-phosphoribosyl-1-pyrophosphate.

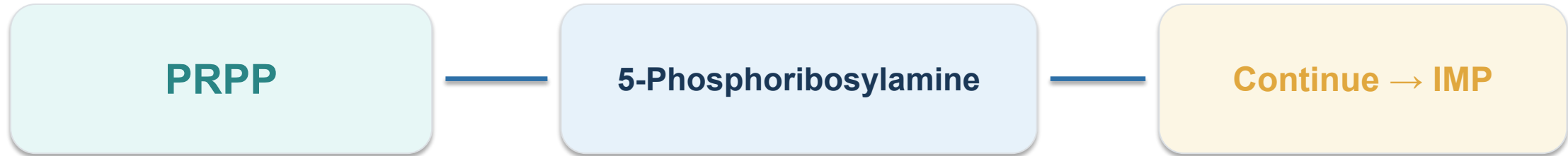


- Enzyme: **PRPP synthetase**.
- Uses ATP; Mg^{2+} is involved.
- Activated by inorganic phosphate (P_i).
- Inhibited by purine ribonucleotides (feedback inhibition).
- PRPP is also used in salvage pathways, so it is not itself the committed step.



Committed step: 5-phosphoribosylamine

This is the first committed step of de novo purine synthesis.

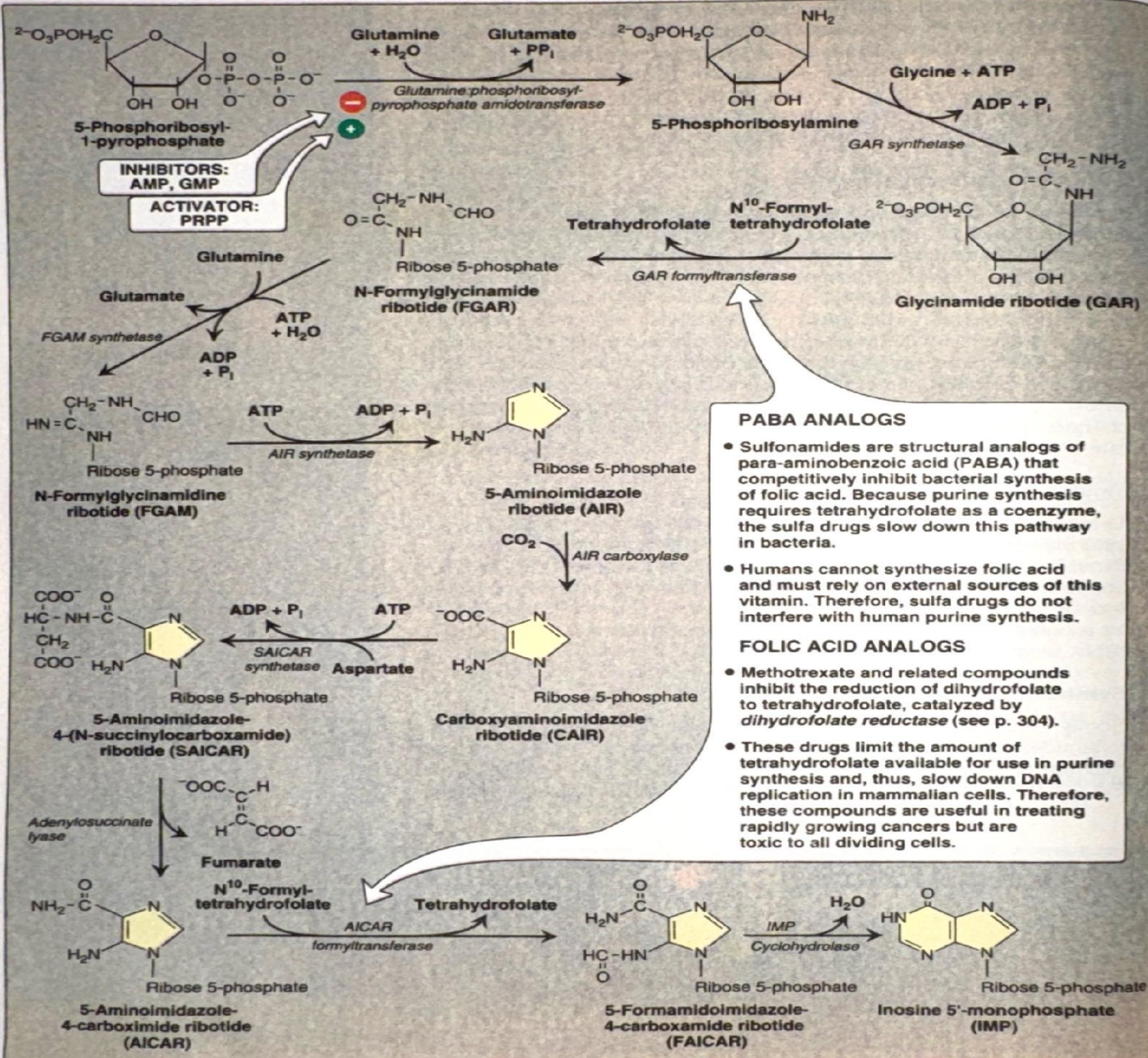


What happens?

- Glutamine donates the amide nitrogen.
- Pyrophosphate is replaced by the glutamine-derived nitrogen.
- Enzyme: glutamine:phosphoribosylpyrophosphate amidotransferase (GPAT).

How is it controlled?

- Inhibited by AMP and GMP (end products).
- Activated by PRPP.
- This is a major control point of de novo purine synthesis.



1. Phosphoribosylamine reacts with **glycine** in the presence of ATP to form glycinamide ribosyl 5-phosphate or glycinamide ribotide (**GAR**).

□ Catalyzed by **GAR synthetase**.

2. **N¹⁰-Formyl tetrahydrofolate** donates the formyl group & the product formed is N - formylglycinamide ribosyl 5-phosphate.

□ The reaction is catalyzed by **formyltransferase**.

3. **Glutamine** transfers the second amido amino group to produce formylglycinamidine ribosyl 5-phosphate.

□ The reaction is catalyzed by **FGAMsynthetase**.

4. The imidazole ring of the purine is closed in an ATP dependent reaction to yield 5- aminoimidazole ribosyl 5-phosphate.

- The reaction is catalyzed by synthetase.

5. Incorporation of CO₂ (carboxylation) occurs to yield aminoimidazole carboxylate ribosyl 5-phosphate.

- The reaction is catalyzed by carboxylase.

6. Does not require the vitamin biotin or ATP.

- Aspartate condenses with the aminoimidazole carboxylate ribosyl 5- phosphate to form aminoimidazole 4- succinylcarboxamide ribosyl 5-phosphate.

- The reaction is catalyzed by synthetase.

7. Adenosuccinate lyase cleaves off fumarate & only the amino group of aspartate is retained to yield aminoimidazole 4-carboxamide ribosyl 5-phosphate.

8. **N¹⁰-Formyl tetrahydrofolate** donates a one- carbon moiety to produce 5- formaminoimidazole 4-carboxamide ribosyl 5- phosphate.

□ **Catalyzed by formyltransferase.**

9. The **final reaction catalyzed by cyclohydrolase** leads to **ring closure** with an **elimination of water molecule**.

□

10. The **product obtained is inosine monophosphate (IMP)**, the parent purine nucleotide from which other purine nucleotides can be synthesized.

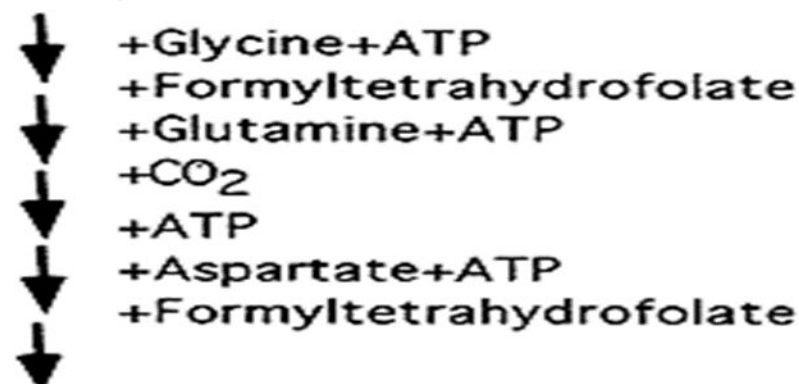
Ribose-5-phosphate (from pentose pathway)



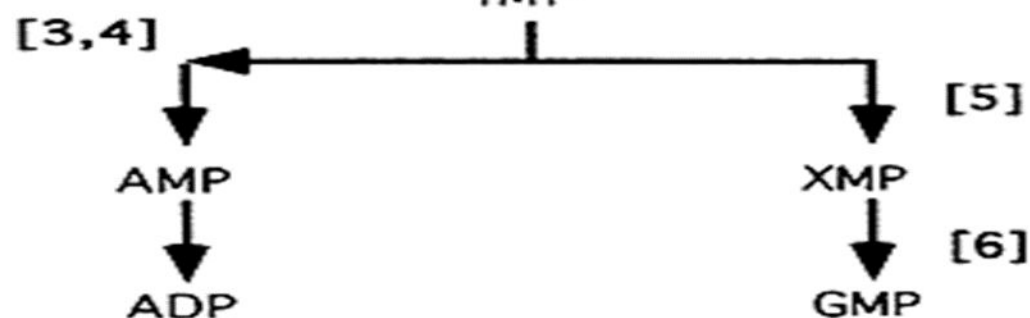
5-Phosphoribosyl-1-pyrophosphate (PRPP)



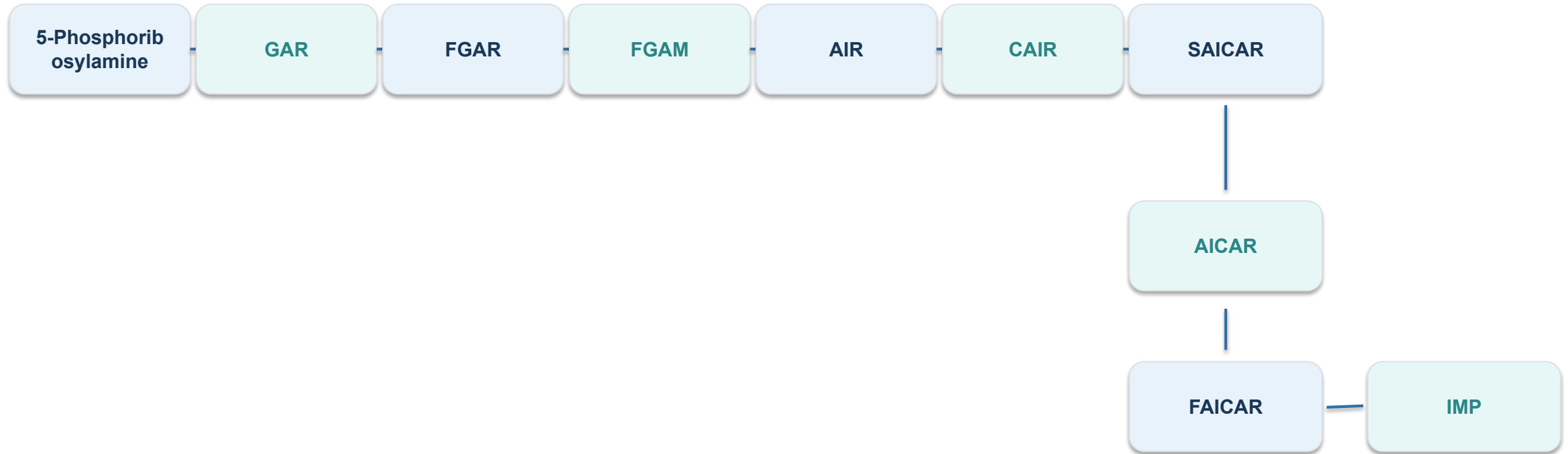
5-Phosphoribosyl amine



IMP



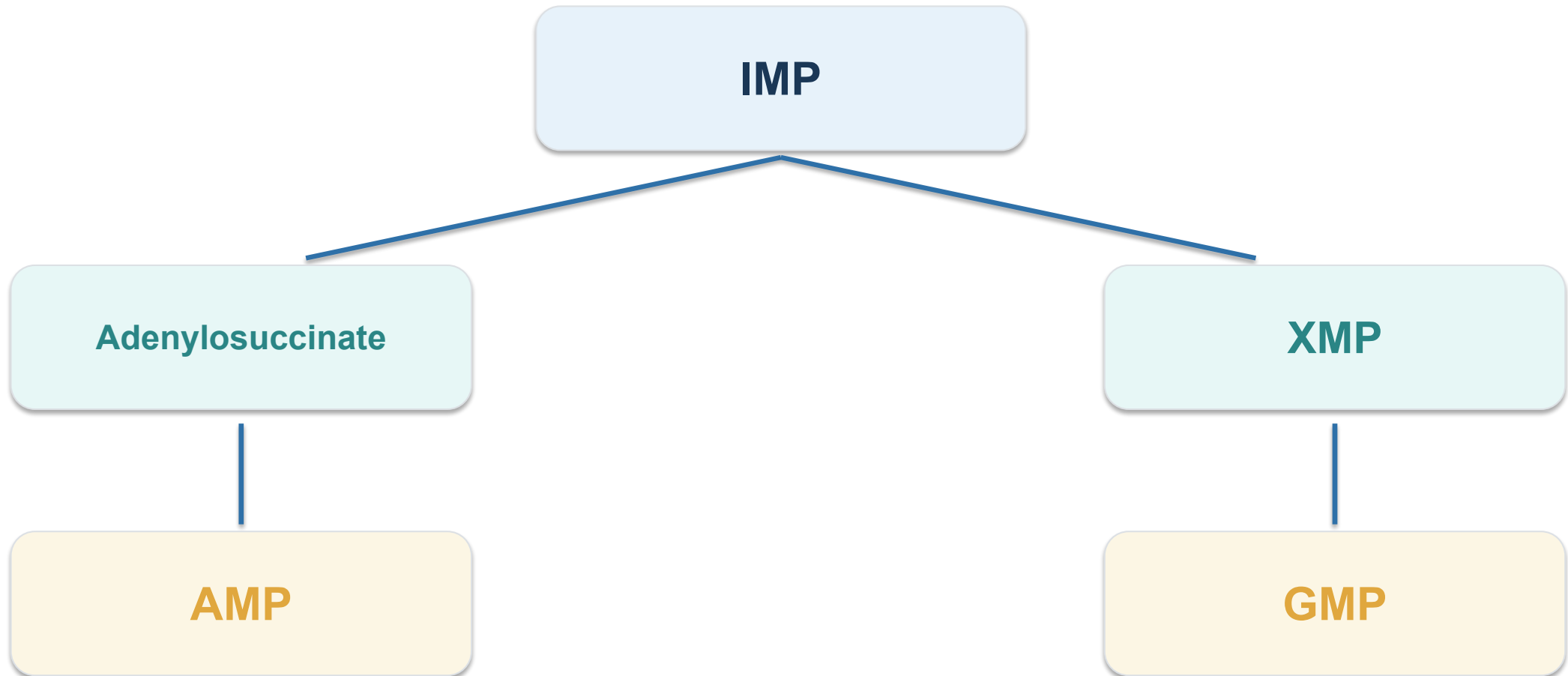
De novo purine pathway: simplified



Important inputs along the pathway: ATP, glutamine, glycine, aspartate, CO₂, and N¹⁰-formyl-THF.

IMP → AMP or GMP

IMP is the common starting point for the two major purine nucleotides.



AMP branch uses GTP + aspartate. GMP branch uses ATP + glutamine. Each branch is inhibited by its own end product, helping balance AMP and GMP.

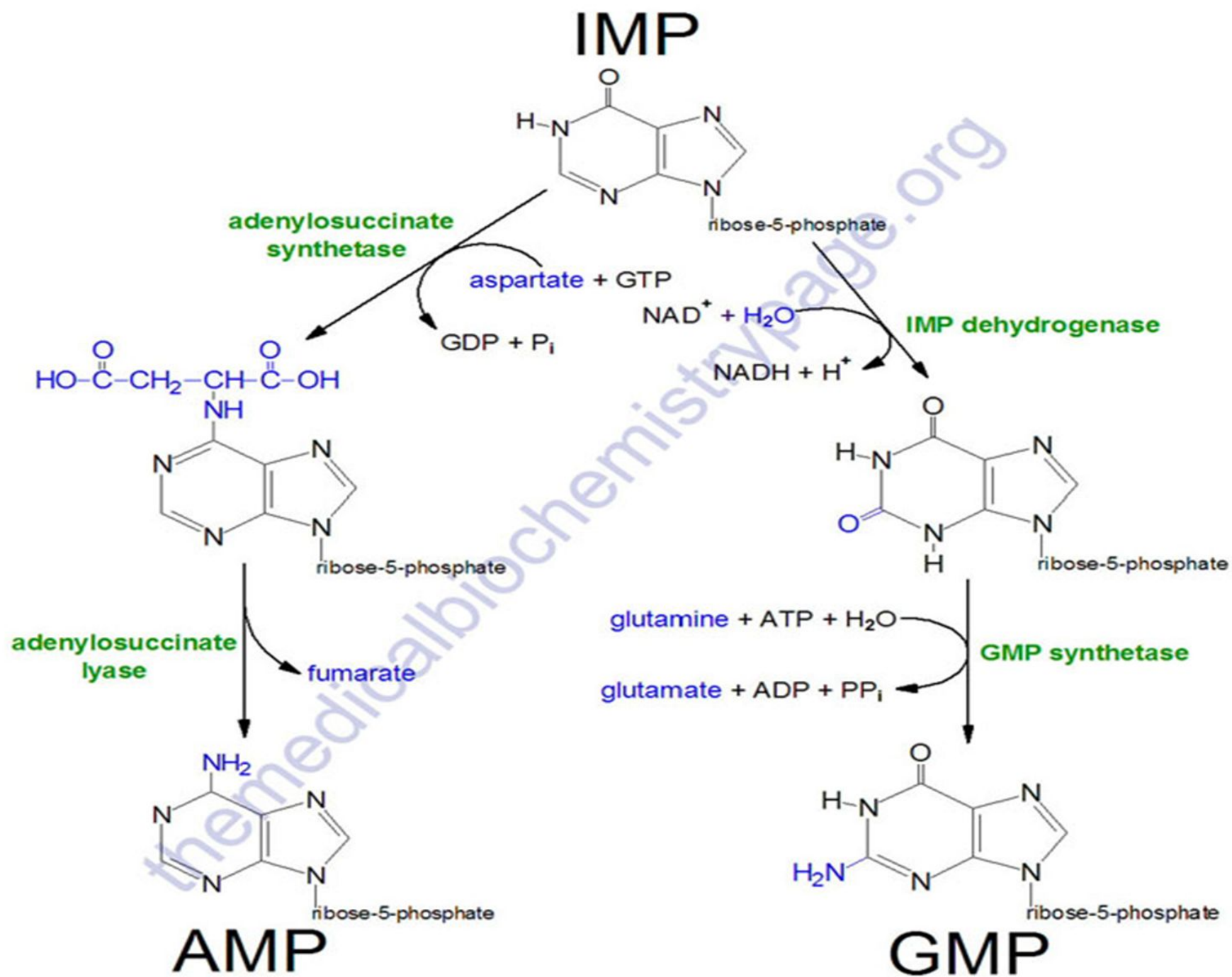
Synthesis of AMP & GMP from IMP

Synthesis of AMP:

- Inosine monophosphate (IMP) is the immediate precursor for the formation of AMP & GMP.
- Aspartate condenses with IMP in the presence of GTP to produce adenylosuccinate which, on cleavage, forms AMP.

Synthesis of GMP:

- IMP undergoes NAD⁺ dependent dehydrogenation to form xanthosine monophosphate (XMP).
- Glutamine then transfers amide nitrogen to xanthosine monophosphate (XMP) to produce GMP.
- 6-Mercaptopurine is an inhibitor of the synthesis of AMP & GMP.
- It acts on the enzyme adenylosuccinase (of AMP pathway).
- IMP dehydrogenase (of GMP pathway).



Drug inhibition of purine synthesis

Some drugs target nucleotide production because rapidly dividing cells need many nucleotides.

Sulfonamides — bacteria

- Sulfonamides are similar in structure to PABA.
- Competitively inhibit bacterial folate synthesis.
- Bacteria need this pathway to make tetrahydrofolate, which is important for their growth and DNA production.
- Humans obtain folate from the diet → sulfonamides do not block human folate synthesis in the same way.

Methotrexate — humans

- Inhibits dihydrofolate reductase (DHFR).
- Reduces tetrahydrofolate available for purine synthesis.
- This slows DNA replication.
- Useful against rapidly growing cancers, but also affects normal dividing cells.

Why toxicity occurs: bone marrow, GI lining, immune cells, and hair follicles normally divide rapidly, so nucleotide synthesis inhibitors can affect them.

- Possible effects: anemia, GI disturbance, immune suppression, and hair loss.

From monophosphate to triphosphate

Cells interconvert nucleotide phosphate levels to match energy and DNA/RNA needs.



Nucleoside diphosphate kinase
(GDP $\leftrightarrow$ GTP; CDP $\leftrightarrow$ CTP)

Base-specific kinases

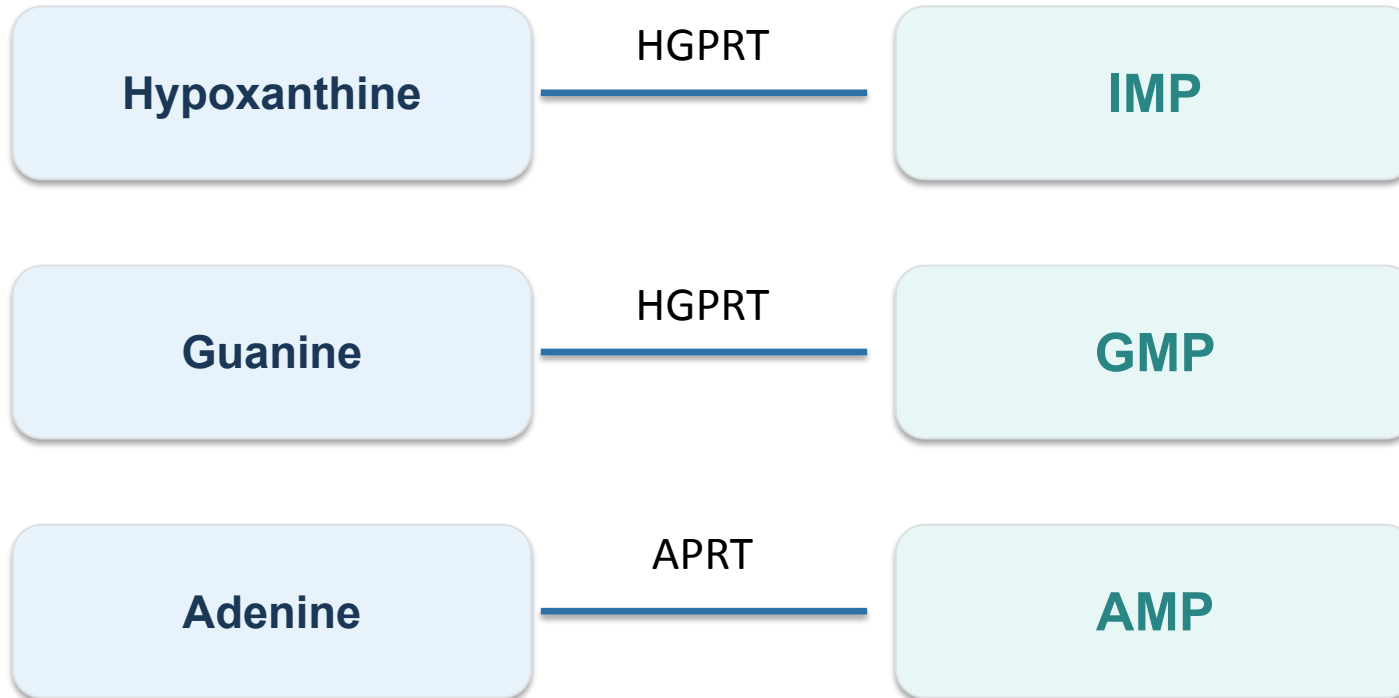
- $\text{AMP} + \text{ATP} \rightarrow 2 \text{ADP}$; $\text{GMP} + \text{ATP} \rightarrow \text{GDP} + \text{ADP}$

Key point

- ATP is generally the main phosphate donor because it is present at high concentration.

Purine salvage pathway

Salvage = recycle purine bases instead of building them from scratch.



PRPP supplies the ribose-phosphate group.
Enzymes: **HGPRT salvages hypoxanthine and guanine; APRT salvages adenine.**

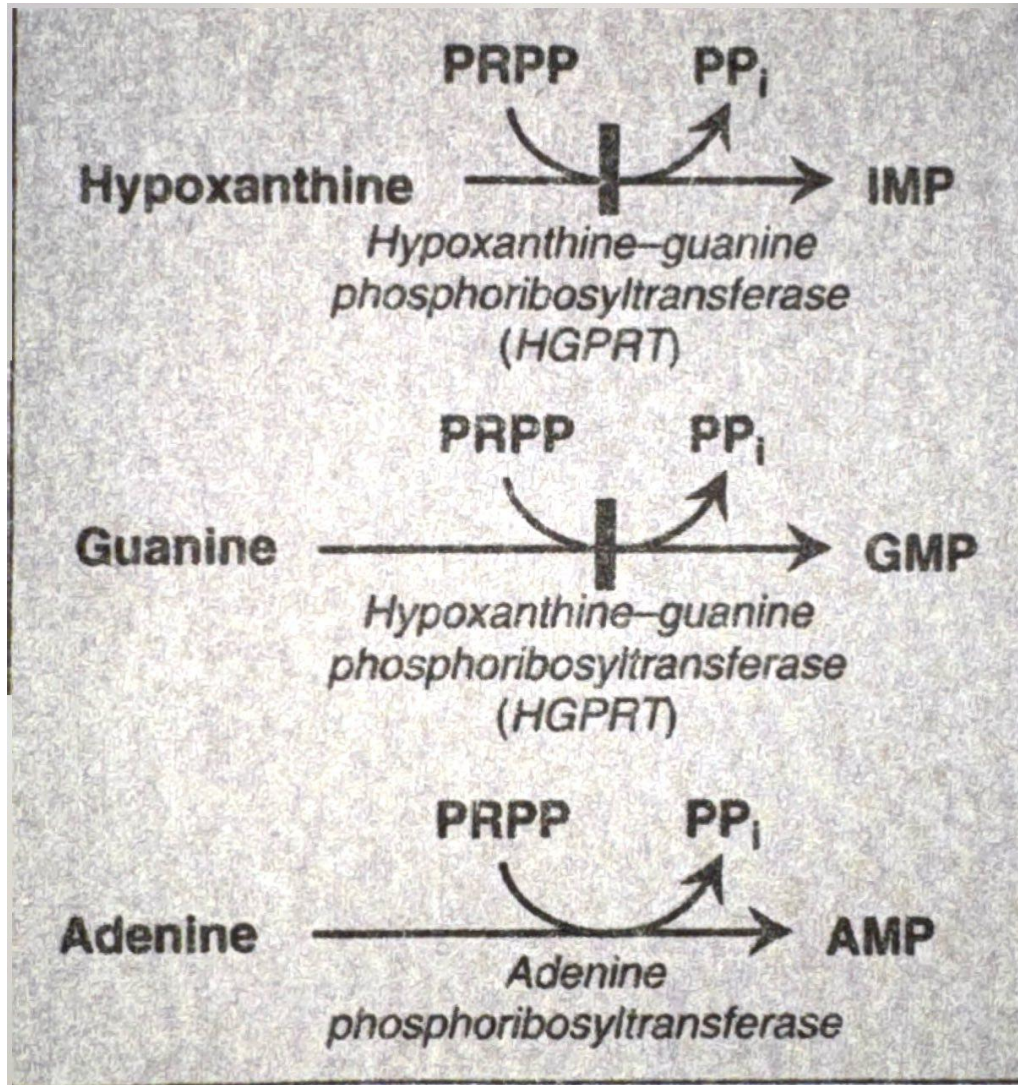
- Salvage is especially important in the brain.
- The reactions release pyrophosphate (PPi), making them effectively irreversible.
- Adenosine is the only purine nucleoside to be salvaged. It can be phosphorylated to AMP by adenosine kinase.

Purine base + PRPP → Purine nucleotide + PPi

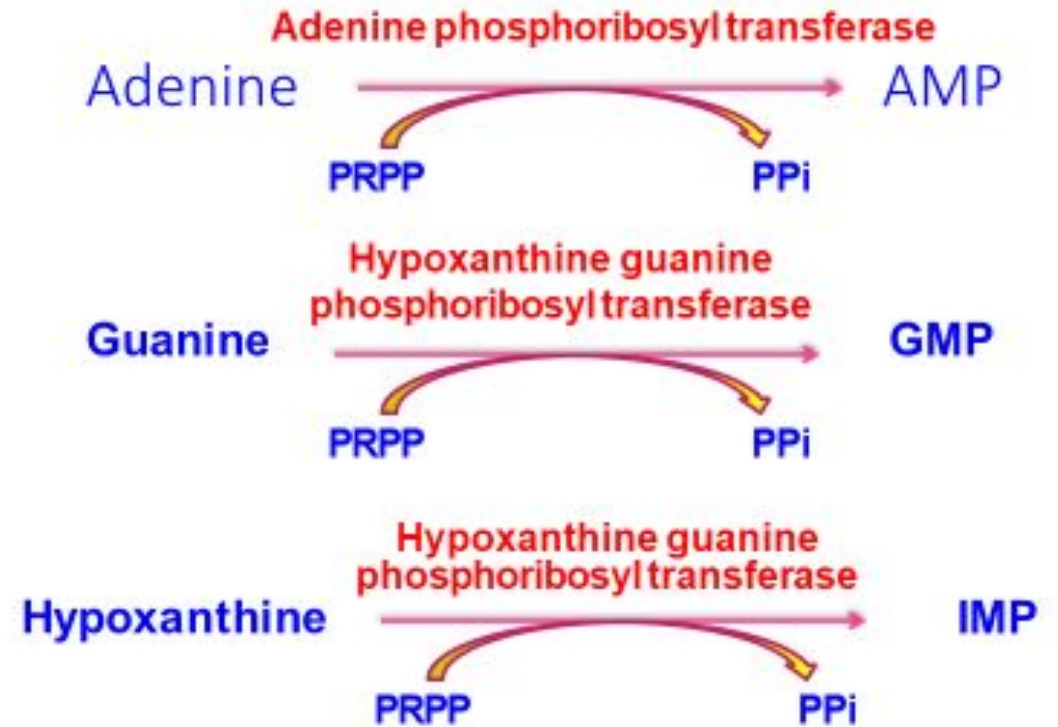
The reaction releases **PPi** as a by-product.

HGPRT= Hypoxanthine-guanine phosphoribosyltransferase

APRT= Adenine phosphoribosyltransferase



Salvage pathway for purines



Lesch–Nyhan syndrome

A classic disorder caused by severe HGPRT deficiency.

Cause

- Rare; X-linked recessive disorder.
- Virtually complete deficiency of HGPRT.
- Hypoxanthine and guanine cannot be efficiently salvaged.

Biochemical effect

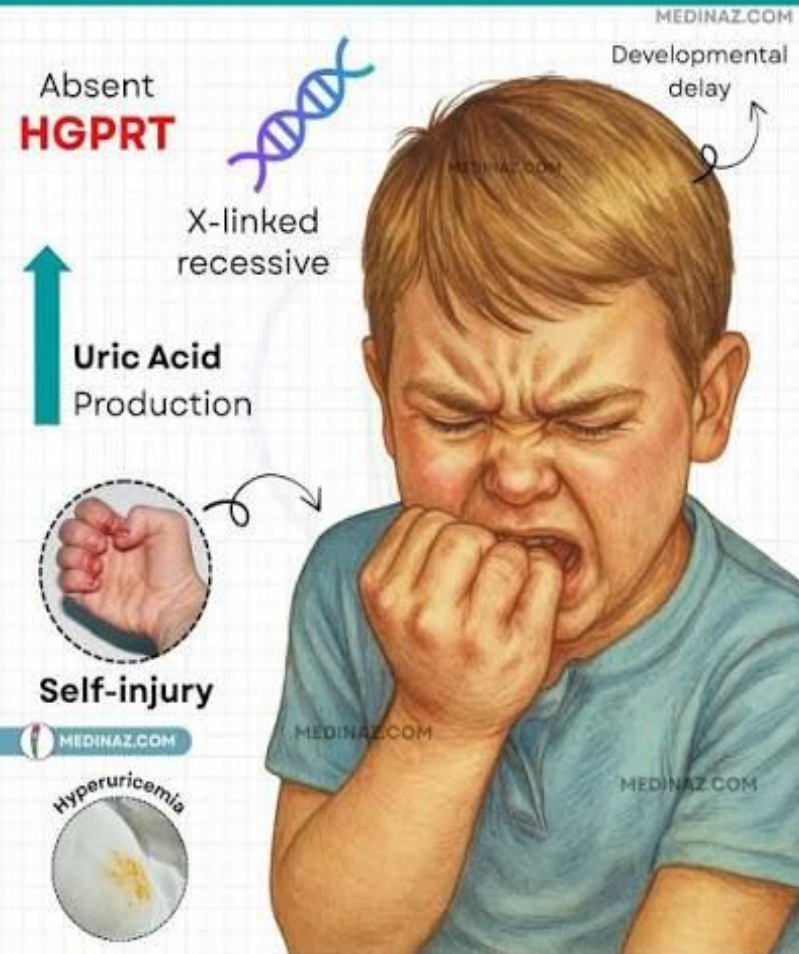
- $\uparrow$ PRPP
- $\downarrow$ IMP and GMP
- GPAT inhibition falls $\rightarrow$ de novo purine synthesis rises.
- More purines are degraded $\rightarrow \uparrow$ uric acid.

Clinical features

- Hyperuricemia
- Kidney stones (urolithiasis)
- Gouty arthritis / urate crystal deposition in joints
- Neurologic and behavioral problems
- Self-injury such as lip/finger biting

Memory link: HGPRT deficiency $\rightarrow$ poor salvage $\rightarrow$ more de novo purine synthesis + more uric acid.

Lesch-Nyhan Syndrome



Lesch-Nyhan syndrome

HGPRT - Defective purine salvage,

↳ ↓ ↓ GMP (from guanine)

Compensatory ↓ ↓ IMP (from hypoxanthine)
purine synthesis

X-linked recessive

EXCESS URIC ACID



Findings: Intellectual disability,

- self-mutilation, aggression,
- hyperuricemia (red/orange "sand" [sodium urate crystals] in diaper),
- gout,
- dystonia (muscle tension),
- macrocytosis.

HGPRT:

- **H** Hyperuricemia
- **G** Gout
- **P** Pissed off (aggression, self-mutilation)
- **R** Red/orange crystals in urine
- **T** Tense muscles (dystonia)

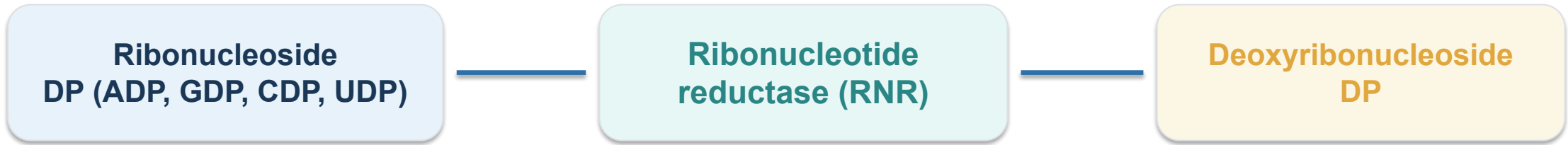
Treatment:

- allopurinol,
- febuxostat



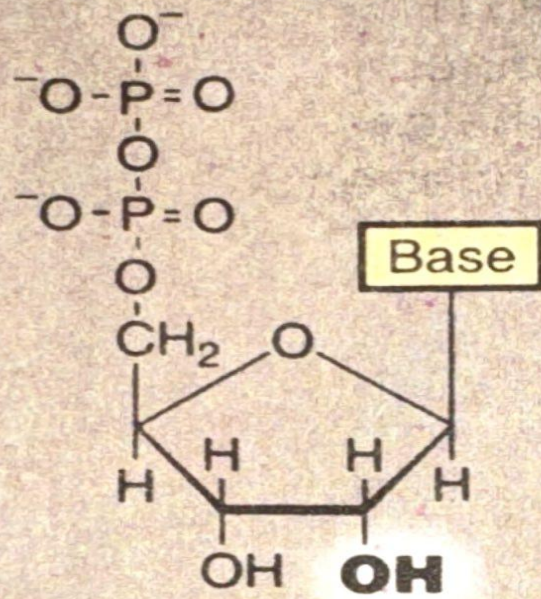
Deoxyribonucleotide synthesis

DNA requires deoxyribonucleotides, which are made by reducing ribonucleotides.

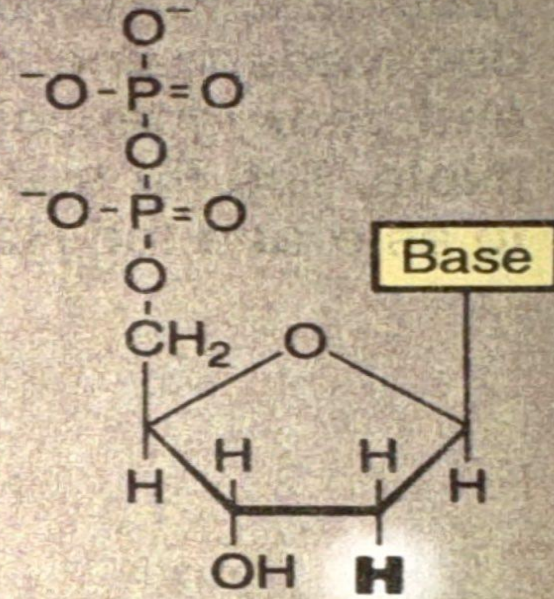


- RNR acts on ribonucleoside diphosphates (NDPs) to form deoxy-NDPs (dNDPs).
- The enzyme has R1 and R2 multisubunit (two B1 & two B2 subunits).
- The reaction removes the 2'-OH group, producing the deoxy form.
- This step is essential for DNA synthesis.

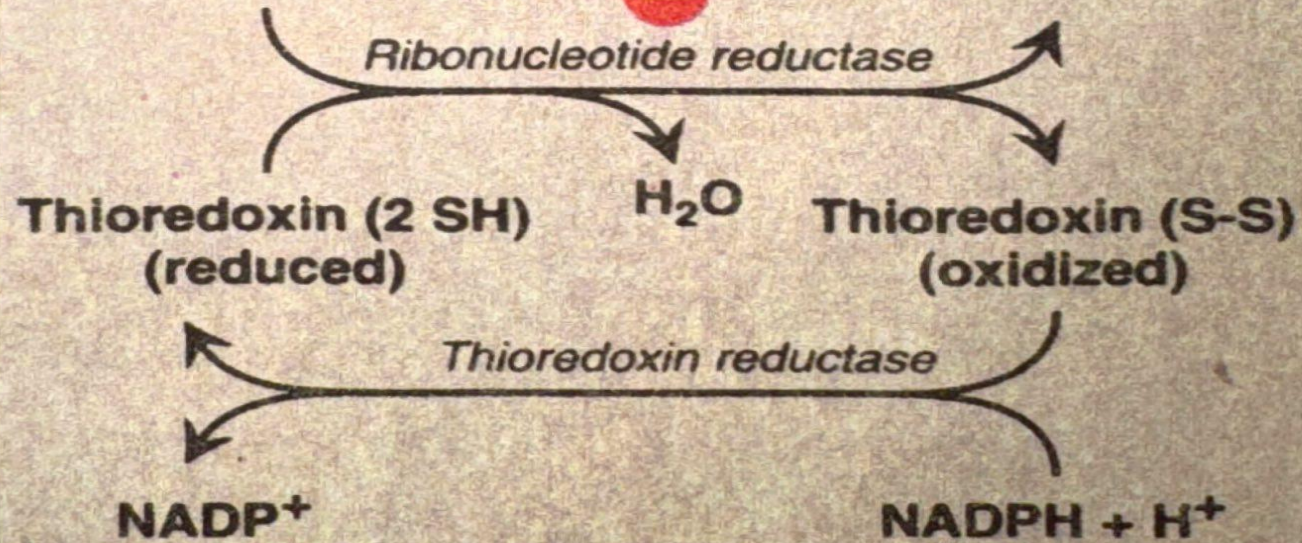
RNR is used during the S phase of the cell cycle, when DNA synthesis is active.



Ribonucleoside diphosphate

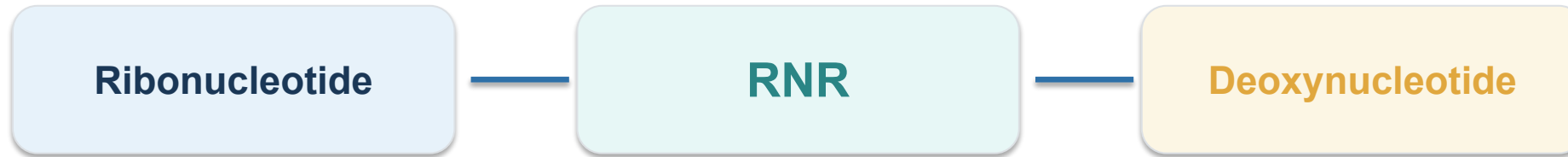


Deoxyribonucleoside diphosphate



How ribonucleotide reductase keeps working

RNR needs a reducing system to restore its active –SH groups.



Step 1 — reduced enzyme

- RNR uses two –SH groups during the reaction.
- They become an oxidized disulfide bond.
- Thioredoxin donates hydrogen atoms to reduce RNR back to its active form.

Step 2 — regenerate thioredoxin

- Thioredoxin must be reduced again.
- NADPH + H⁺ provide the reducing equivalents.
- Thioredoxin reductase performs this regeneration.

Easy chain to remember: NADPH → thioredoxin reductase → reduced thioredoxin → RNR → deoxyribonucleotide.

Regulation of ribonucleotide reductase

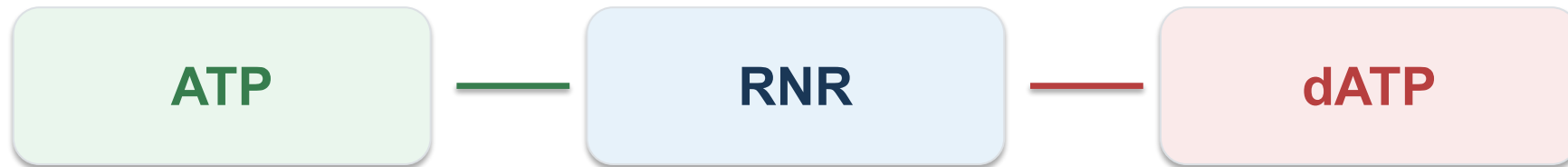
The cell must keep the four dNTPs balanced for accurate DNA synthesis.

Activity site

- ATP binding → activates RNR.
- dATP binding → inhibits overall RNR activity.
- Too much dATP can shut down reduction of all four NDPs.

Substrate-specificity sites

- Binding of ATP, dATP, dTTP, or dGTP changes which NDP is reduced.
- This helps maintain balanced amounts of dATP, dGTP, dCTP, and dTTP.



Think of RNR as the 'dNTP supply manager': ATP turns it on; dATP can turn it off; other nucleotides help choose the correct substrate.

Hydroxyurea: an important RNR inhibitor

A clinically useful drug that blocks deoxyribonucleotide production.

Mechanism

- Hydroxyurea inhibits ribonucleotide reductase.
- It interferes with the enzyme's radical chemistry.
- Less RNR activity → fewer deoxyribonucleotides → less DNA synthesis.

Uses

- Antineoplastic therapy for some cancers, such as melanoma.
- Used in sickle cell anemia.
- Because rapidly dividing cells depend on DNA synthesis, inhibition can be therapeutically useful.

High-yield association: Hydroxyurea → ribonucleotide reductase inhibition → ↓ dNTPs → ↓ DNA synthesis.

- However, the increased fetal hemoglobin seen with hydroxyurea is related to changes in gene expression, not simply RNR inhibition.

High-yield summary

If you remember these points, you have the framework of the chapter.

Nucleotide Base + sugar + phosphate(s)

PRPP Activated ribose; made by PRPP synthetase

IMP Parent purine → AMP or GMP

Lesch–Nyhan HGPRT deficiency → ↑ uric acid
+ self-injury

RNR control ATP activates; dATP inhibits

Purines A + G; ring built on PRPP

Committed step GPAT → 5-phosphoribosylamine

Salvage Recycles bases; HGPRT & APRT

RNR NDP → dNDP for DNA

Hydroxyurea Inhibits RNR → ↓ DNA synthesis

PURINE NUCLEOTIDE DEGRADATION

Purine Nucleotide Degradation

From dietary nucleic acids → purine bases → uric acid

Main idea

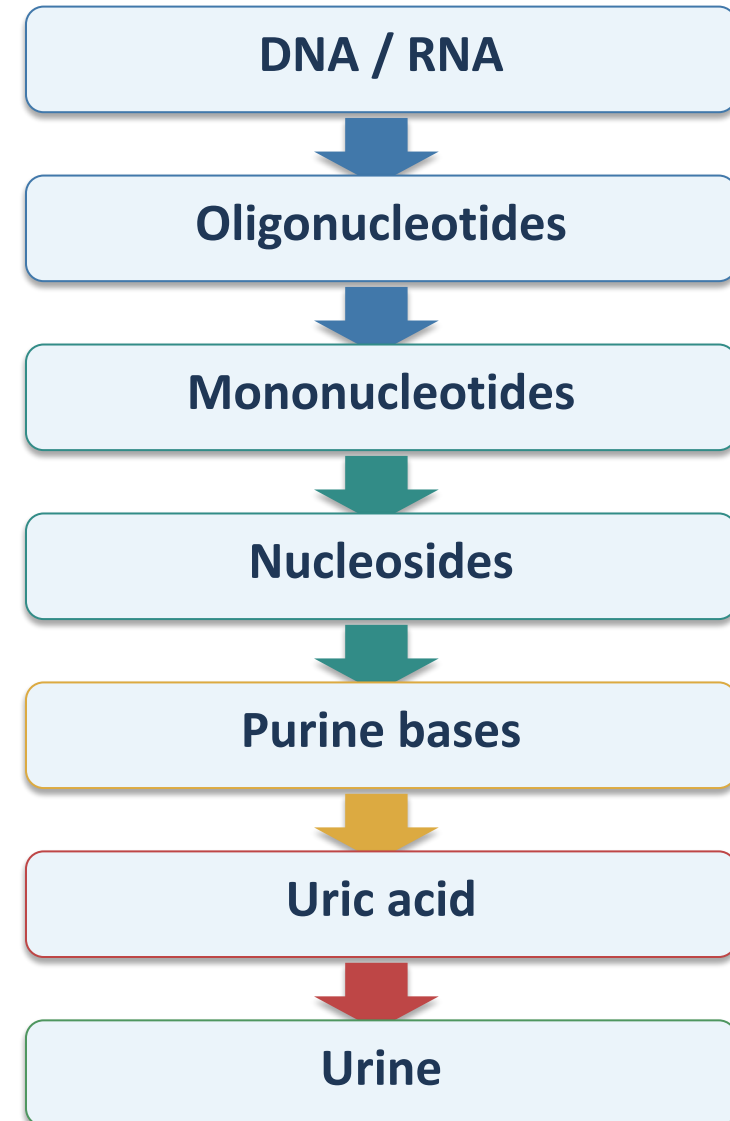
Purines are broken down mainly to uric acid in humans. Too much uric acid can cause hyperuricemia and gout.

Important disorders

Gout
Adenosine deaminase (ADA) deficiency
Purine nucleoside phosphorylase (PNP) deficiency

1. What happens to dietary nucleic acids?

- DNA and RNA from food are digested mainly in the **small intestine**.
- **Pancreatic nucleases** cut DNA/RNA into smaller pieces called **oligonucleotides**.
- **Phosphodiesterases** then make **mono-nucleotides**.
- **Nucleotidases** remove phosphate → **nucleosides**.
- **Nucleosidases / nucleoside phosphorylases** release free bases.
- **Purine bases** are not used much to make tissue nucleic acids; they are mainly degraded to **uric acid**.
- Most uric acid enters the **blood** and is eventually excreted in **urine**.



2. Degradation in the small intestine

Easy way to remember: cut → remove phosphate → release base

Step 1 — Cut DNA & RNA

Pancreatic ribonucleases and deoxyribonucleases break dietary RNA and DNA into oligonucleotides.

Step 2 — Make nucleotides

Pancreatic phosphodiesterases produce 3'- and 5'-mononucleotides.

Step 3 — Remove phosphate

Intestinal nucleotidases remove phosphate groups and form nucleosides.

Step 4 — Release bases

Nucleosidases / nucleoside phosphorylases split nucleosides into free bases + (deoxy)ribose-1-phosphate.

Step 5 — Final fate

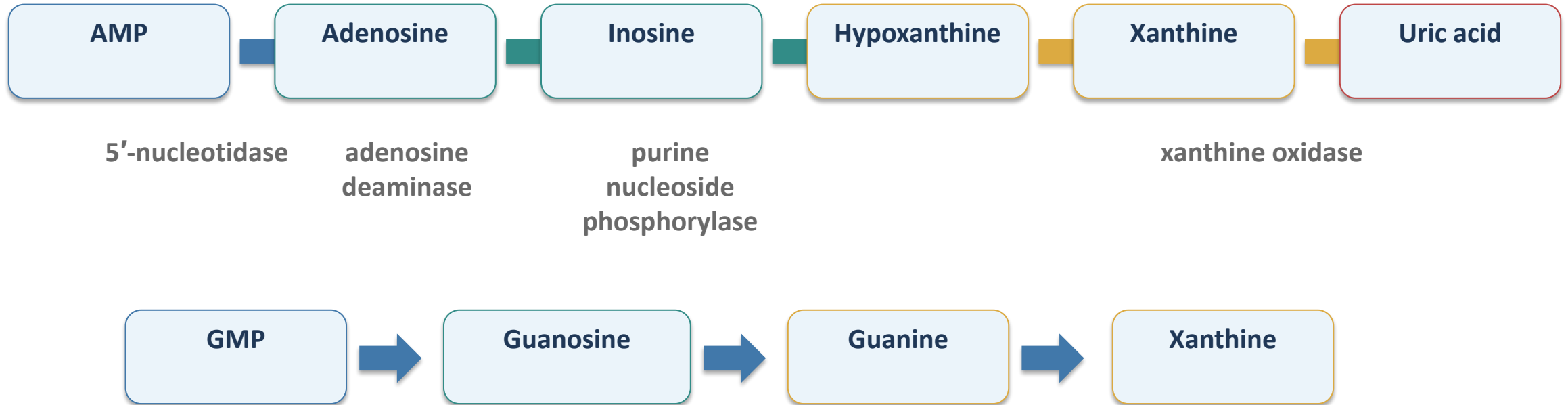
Dietary purine bases are degraded inside enterocytes to uric acid. Uric acid then enters blood and is excreted in urine.

Key note

In mammals other than primates, urate oxidase (uricase) converts urate to allantoin. Humans do not have active uricase.

3. Purine degradation pathway

The two major purines are AMP and GMP

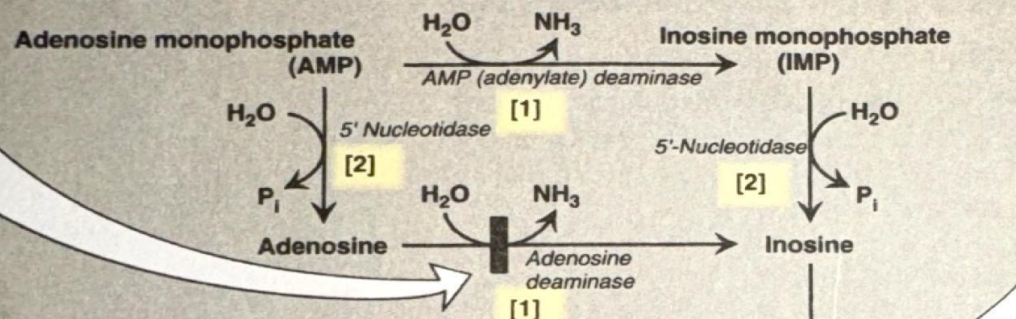


AMP → IMP/inosine → hypoxanthine → xanthine → uric acid

GMP → guanosine → guanine → xanthine → uric acid

ADENOSINE DEAMINASE (ADA) DEFICIENCY

- This autosomal-recessive deficiency causes a type of severe combined immunodeficiency (SCID), involving T-cell, B-cell, and natural killer-cell depletion (lymphocytopenia).
- Untreated ADA-deficient children usually die before age 2 years from overwhelming infection; treatments include BMT, ERT, and gene therapy.

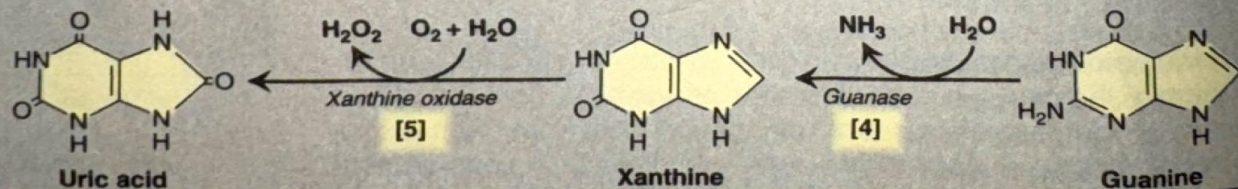


PURINE NUCLEOSIDE PHOSPHORYLASE (PNP) DEFICIENCY

- This autosomal-recessive deficiency is rarer and less severe than ADA deficiency.
- It affects T cell development, primarily.
- PNP-deficient individuals have recurrent infections and neurodevelopmental delay.

GOUT

- This disorder is characterized by hyperuricemia with recurrent attacks of acute arthritic joint inflammation, caused by deposition of monosodium urate crystals.
- In gout, the hyperuricemia results primarily from the underexcretion of uric acid. Overproduction of uric acid is less common, and known causes involve certain inborn errors of metabolism or increased availability of purines.
- Crystal deposition (tophi) may be seen in soft tissues and in the kidneys (urolithiasis).
- Treatment with allopurinol inhibits *xanthine oxidase*, resulting in accumulation of hypoxanthine and xanthine, compounds more soluble than uric acid.



4. Important enzyme steps

What each enzyme does

1. AMP deaminase / Adenosine deaminase

Removes an amino group. AMP $\rightarrow$ IMP; adenosine $\rightarrow$ inosine.

2. 5'-Nucleotidase

Removes phosphate. IMP $\rightarrow$ inosine; GMP $\rightarrow$ guanosine.

3. Purine nucleoside phosphorylase (PNP)

Breaks inosine and guanosine into hypoxanthine and guanine + ribose-1-phosphate.

4. Guanase

Removes an amino group from guanine $\rightarrow$ xanthine.

5. Xanthine oxidase (XO)

Oxidizes hypoxanthine $\rightarrow$ xanthine $\rightarrow$ uric acid.

Key point

Xanthine oxidase is the key enzyme targeted by **allopurinol** and **febuxostat** to reduce uric acid formation.

5. Gout — when uric acid becomes a problem

Cause

Gout is linked to high blood uric acid (hyperuricemia). This is usually because the kidneys do not excrete enough uric acid; overproduction is less common.

What happens?

Uric acid can form monosodium urate (MSU) crystals. These crystals deposit in joints and trigger strong inflammation.

Symptoms / complications

- Recurrent acute attacks of painful joint inflammation
- Chronic gout may develop
- Tophi = crystal masses in soft tissues
- Uric acid kidney stones (urolithiasis) may occur

Diagnosis

Fluid is aspirated from the swollen joint (arthrocentesis) and examined under polarized light. **Needle-shaped, negatively birefringent MSU crystals** confirm **gout**.

Remember

Hyperuricemia can be asymptomatic. It is necessary before gout, but high uric acid alone does not mean a person has gout.

6. Why does hyperuricemia happen?

A. Under-excretion — most common

More than 90% of people with hyperuricemia have under-excretion of uric acid.

- Kidney handling defects (primary or inherited).
- **Secondary causes:** diseases that change kidney urate handling.
- **Lactic acidosis** can increase renal urate reabsorption.
- Some drugs, such as **thiazide diuretics**, reduce uric acid excretion.
- **Lead exposure** can cause saturnine gout.

B. Overproduction — less common

More uric acid is produced because purine breakdown is increased or purine synthesis is abnormally high.

- Usually idiopathic (no clear cause).
- Certain PRPP synthetase mutations can increase purine production.
- Reduced purine salvage, as in Lesch–Nyhan syndrome, can increase PRPP and purine synthesis.
- Secondary overproduction can occur with rapid cell turnover, e.g., some myeloproliferative disorders or chemotherapy.

7. Gout treatment

Acute attack vs. long-term prevention

Acute attack — reduce inflammation

- Colchicine
- NSAIDs such as indomethacin
- Steroids such as prednisone

These control inflammation and pain. Colchicine reduces movement of neutrophils into the affected area.

Allopurinol — simple idea

Allopurinol is converted to **oxypurinol**, a long-lasting **XO inhibitor**. **Hypoxanthine** and **xanthine** accumulate; they are more soluble than uric acid.

Long-term goal

Keep uric acid below its **saturation point** (~6.5 mg/dL) so MSU crystals do not keep forming.

If uric acid excretion is low

Uricosuric drugs such as **probenecid** or **sulfinpyrazone** increase renal uric acid excretion.

If uric acid production is high

Allopurinol inhibits **xanthine oxidase**. **Febuxostat** is another non-purine xanthine oxidase inhibitor.

8. Diet and gout risk

Higher risk

A diet rich in meat, seafood—especially shellfish—and ethanol (alcohol) is associated with a higher risk of gout.

Lower risk

A diet richer in low-fat dairy products is associated with a lower risk of gout.

Why it matters

Diet can influence uric acid levels and the risk of gout attacks, but gout is also strongly affected by kidney uric acid handling and genetics.

9. Adenosine deaminase (ADA) deficiency

What is ADA?

ADA is a **cytoplasmic enzyme** found in many tissues. Lymphocytes have especially high ADA activity.

What happens when ADA is missing?

Adenosine accumulates and is converted to ribonucleotides/deoxyribonucleotides. This raises dATP, which inhibits ribonucleotide reductase.

Why are lymphocytes affected?

Ribonucleotide reductase inhibition blocks production of deoxyribonucleotides needed for DNA synthesis. Cells cannot divide normally → lymphocyte loss.

Clinical result

Severe ADA deficiency causes autosomal-recessive **SCID** (severe combined immunodeficiency), with very **low T cells, B cells and natural killer cells**.

Treatment

Bone marrow transplantation, enzyme replacement therapy, and gene therapy may be used. Without treatment, severe cases can lead to death from infection in early childhood.

10. Purine nucleoside phosphorylase (PNP) deficiency

What does PNP do?

PNP converts inosine and guanosine into the **purine bases hypoxanthine and guanine**, plus **ribose-1-phosphate**.

Deficiency

PNP deficiency is an **autosomal-recessive disorder**. It is rarer and usually **less severe than ADA deficiency**.

Main effects

T-cell development is mainly affected. Patients may have **recurrent infections** and **neurodevelopmental delay**.

Quick comparison

ADA deficiency → severe combined immunodeficiency (T, B and NK cells affected).

PNP deficiency → mainly T-cell dysfunction; generally milder.

11. Drug connection: Hydroxyurea

Mechanism

Hydroxyurea (hydroxycarbamide) inhibits **ribonucleotide reductase**.

Result

Less ribonucleotide →  deoxyribonucleotide conversion → fewer substrates available for DNA synthesis.

Clinical use

It is an antineoplastic drug used in cancers such as melanoma and is also used in sickle cell anemia.

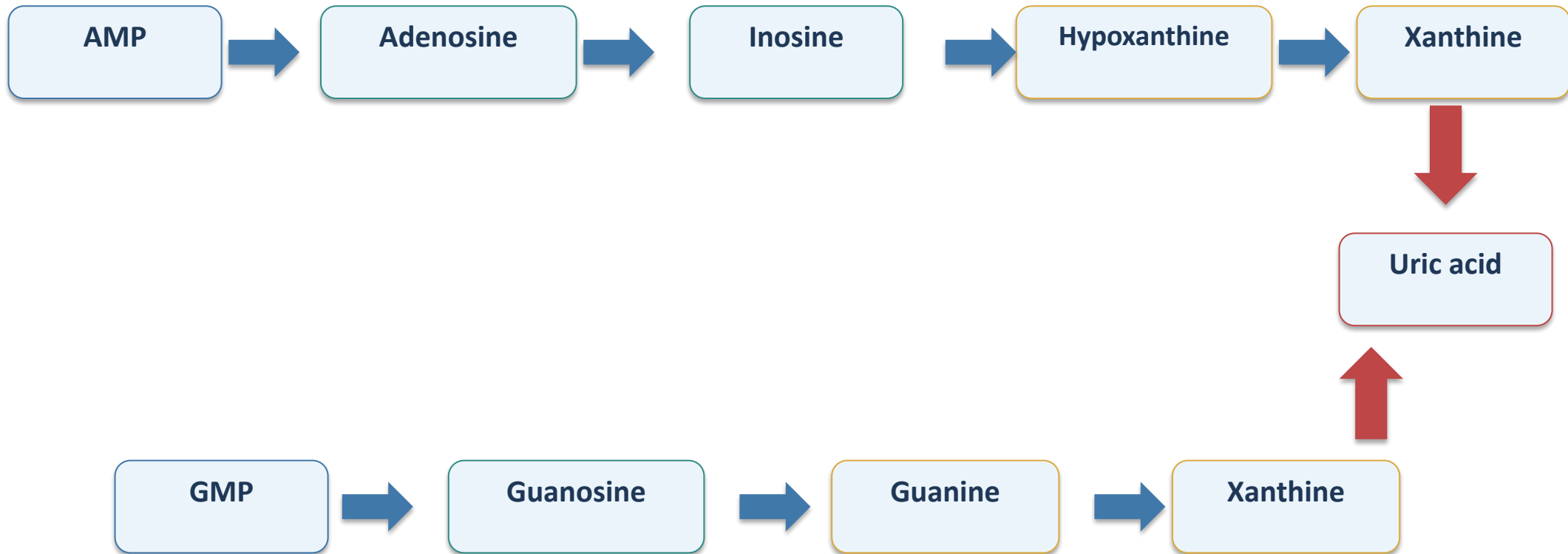
Important note

The rise in fetal hemoglobin (HbF) seen with hydroxyurea in sickle cell disease is mainly due to changes in gene expression, not simply ribonucleotide reductase inhibition.

Link to ADA deficiency

Both topics involve ribonucleotide reductase, but in ADA deficiency it is inhibited indirectly by **accumulated dATP**.

12. High-yield pathway to memorize



Key points:

AMP → inosine → hypoxanthine → xanthine → uric acid

GMP → guanosine → guanine → xanthine → uric acid

13. Final high-yield summary

Purine end product

In humans, purine degradation ends mainly in uric acid.

Gout

Usually due to under-excretion of uric acid → MSU crystal deposition → inflammation.

Diagnosis

Joint fluid + polarized microscopy → needle-shaped, negatively birefringent MSU crystals.

Treatment

Acute: colchicine/NSAIDs/steroids. Long-term: lower uric acid with uricosurics or XO inhibitors.

ADA deficiency

↑ adenosine → ↑ dATP → inhibits ribonucleotide reductase → impaired DNA synthesis → SCID.

PNP deficiency

Mainly affects T cells; recurrent infections and neurodevelopmental problems can occur.

XO inhibitors

Allopurinol and febuxostat decrease uric acid formation.

THANK YOU

FOR WATCHING MY PRESENTATION

